ASSIGNMENT

PROGRMMING FUNDAMENTAL LAB 2:

TASK 01:

O¹:

```
# include <iostream>
using namespace std;
int main ( )
{
   char letter;
   letter = 'A';
cout <<letter<< '\n';
   letter = 'B';
cout <<letter <<'\n';
   return 0;
}
```

OUTPUT:

```
A
B
```

O²

```
# include <iostream>
using namespace std;
int main ( )
{
  char letter;
  letter = 65;
cout <<letter << '\n';
   letter = 66;
cout <<letter <<'\n';
  cin >> letter;
  return 0;
}
```

OUTPUT:

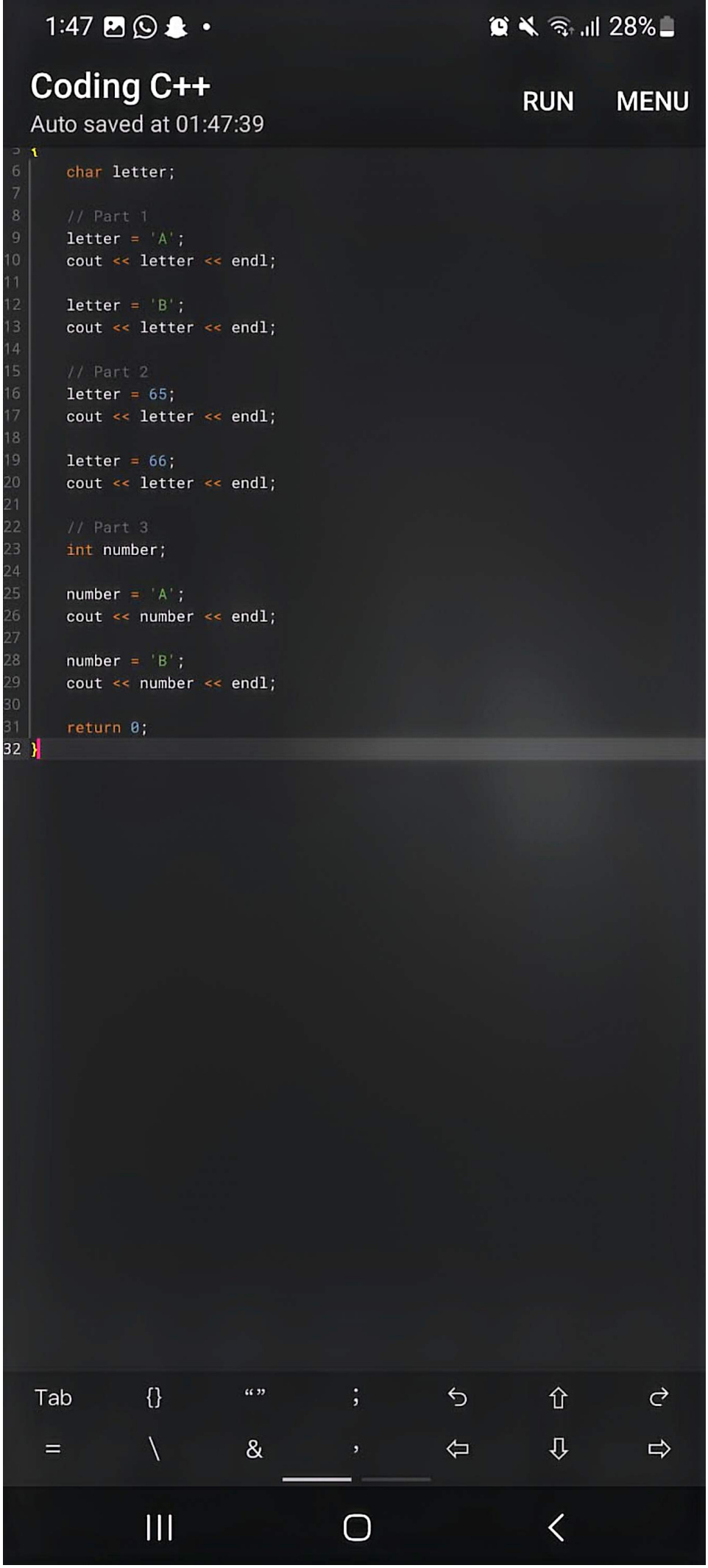
```
A
B
```

O³:

```
#include <iostream>
using namespace std;
int main ( )
{
   int letter;
  Letter = 'A';
cout <<letter<< '\n';
  Letter = 'B';
cout << Letter <<'\n';
  cin >> letter;
   return 0;
}
```

OUTPUT

```
65
66
```


1:47
28%
Coding C++
Auto saved at 01:47:39
RUN
MENU
char letter;
// Part 1
letter = 'A';
cout << letter << endl;
letter = 'B';
cout << letter << endl;
// Part 2
letter = 65;
cout << letter << endl;
letter = 66;
cout << letter << endl;
// Part 3
int number;
number = 'A';
cout << number << endl;
number = 'B';
cout << number << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner


1:47
28%
Compile Result
A
B
A
B
65
66
[Process completed - press Enter]

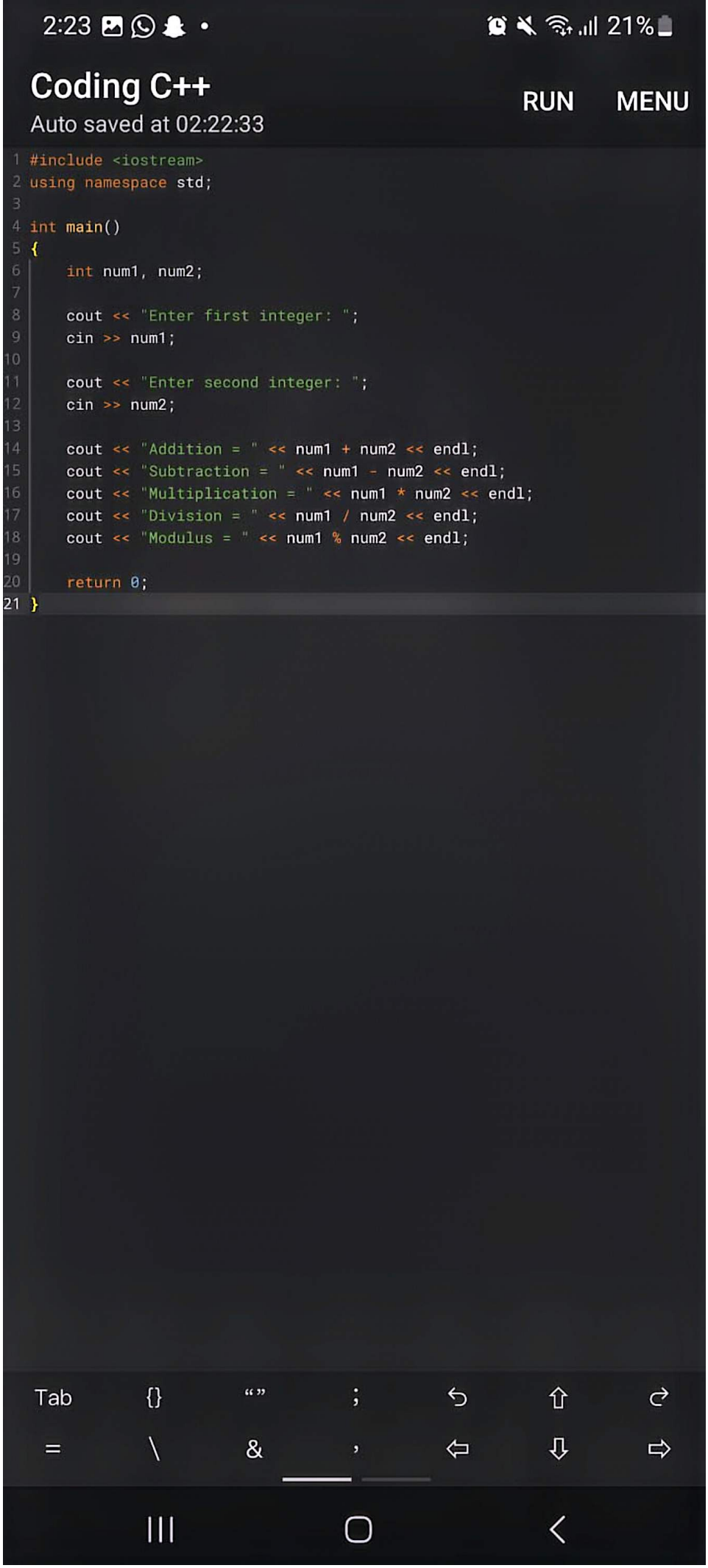
CamScanner

TASK 02:

```
# include <iostream>
using namespace std;
int main ( )
{
  int num 1, num 2
cout << Enter two integers: ";
cin >> num 1 >> num 2;
cout << " Addition = " << num 1 + num 2 << endl;
cout << " Subtraction = " << num 1 - num 2 << endl;
cout << Multiplication = " << num 1 * num 2 << endl;
cout << Division = " << num 1/num 2 << endl;
      return 0;
}
```

OUTPUT:

```
Enter two integers : 20  5
    Addition = 25
    Subtraction = 15
    Multiplication = 100
    Division = 4
```


2:23
21%
Coding C++
Auto saved at 02:22:33
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int num1, num2;
cout << "Enter first integer: ";
cin >> num1;
cout << "Enter second integer: ";
cin >> num2;
cout << "Addition = " << num1 + num2 << endl;
cout << "Subtraction = " << num1 - num2 << endl;
cout << "Multiplication = " << num1 * num2 << endl;
cout << "Division = " << num1 / num2 << endl;
cout << "Modulus = " << num1 % num2 << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

```
Enter first integer: 20
Enter second integer: 5
Addition = 25
Subtraction = 15
Multiplication = 100
Division = 4
Modulus = 0

[Process completed - press Enter]
```

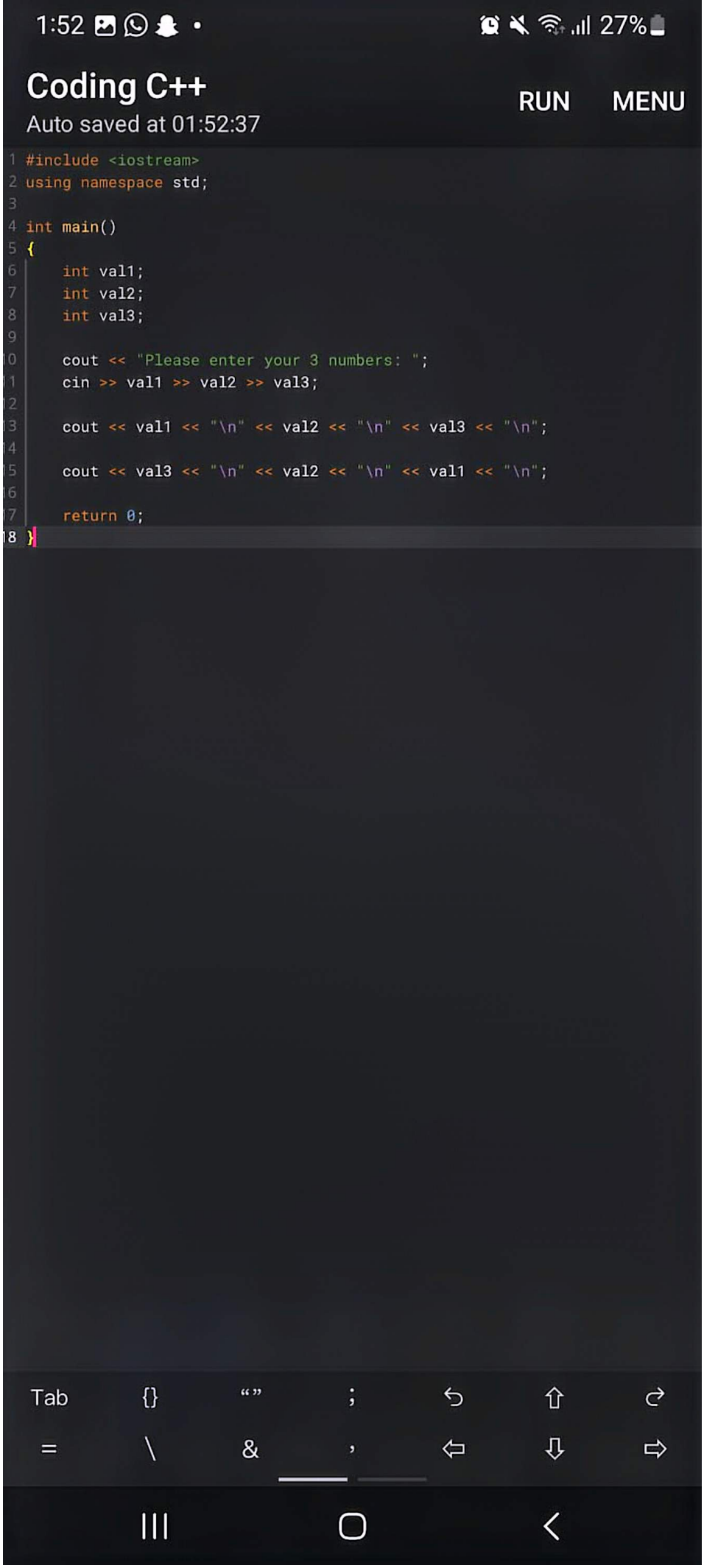
CamScanner

TASK 03:

```
#include "stdafx.h"
#include <iostream>
Using namespace std;
int main ( )
{
    int val 1;
    int val 2;
    int val 3;
Cout << Please enter your 3 numbers: ";
cin >> val 1 >> val 2 >> val 3;
cout << val 1 << " \n " << val 2 << " \n " << val 3 << " \n ";
cout << val 3 << " \n " << val 2 << " \n " << val 1 << " \n ";
    system ("Pause")
    return 0;
}
```

OUTPUT

```
Please enter your 3 numbers
10
20
30
30
20
10
```


1:52
27%
Coding C++
Auto saved at 01:52:37
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int val1;
int val2;
int val3;
cout << "Please enter your 3 numbers: ";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

```
Please enter your 3 numbers: 10 20 30
10
20
30
30
20
10

[Process completed - press Enter]
```

CamScanner

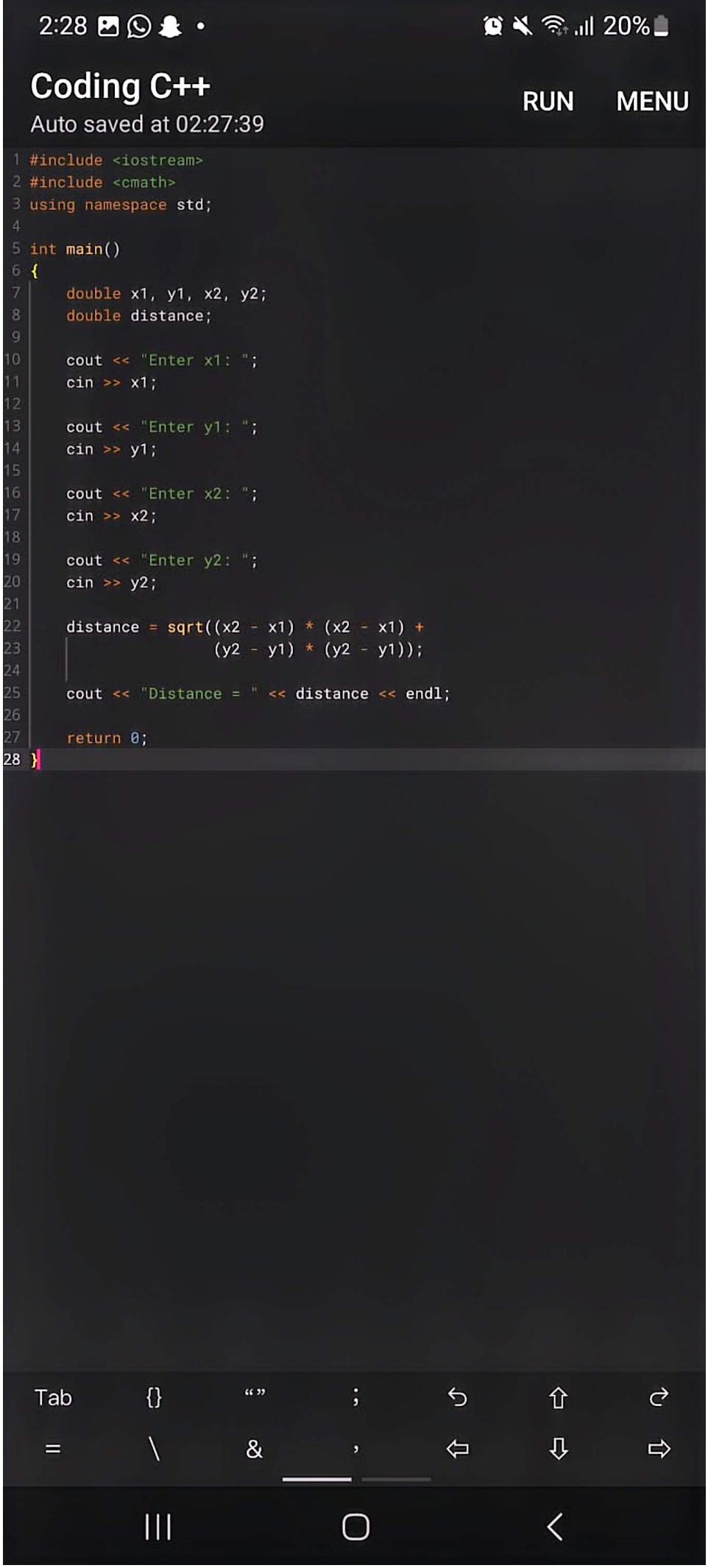
TASK 04:

Formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

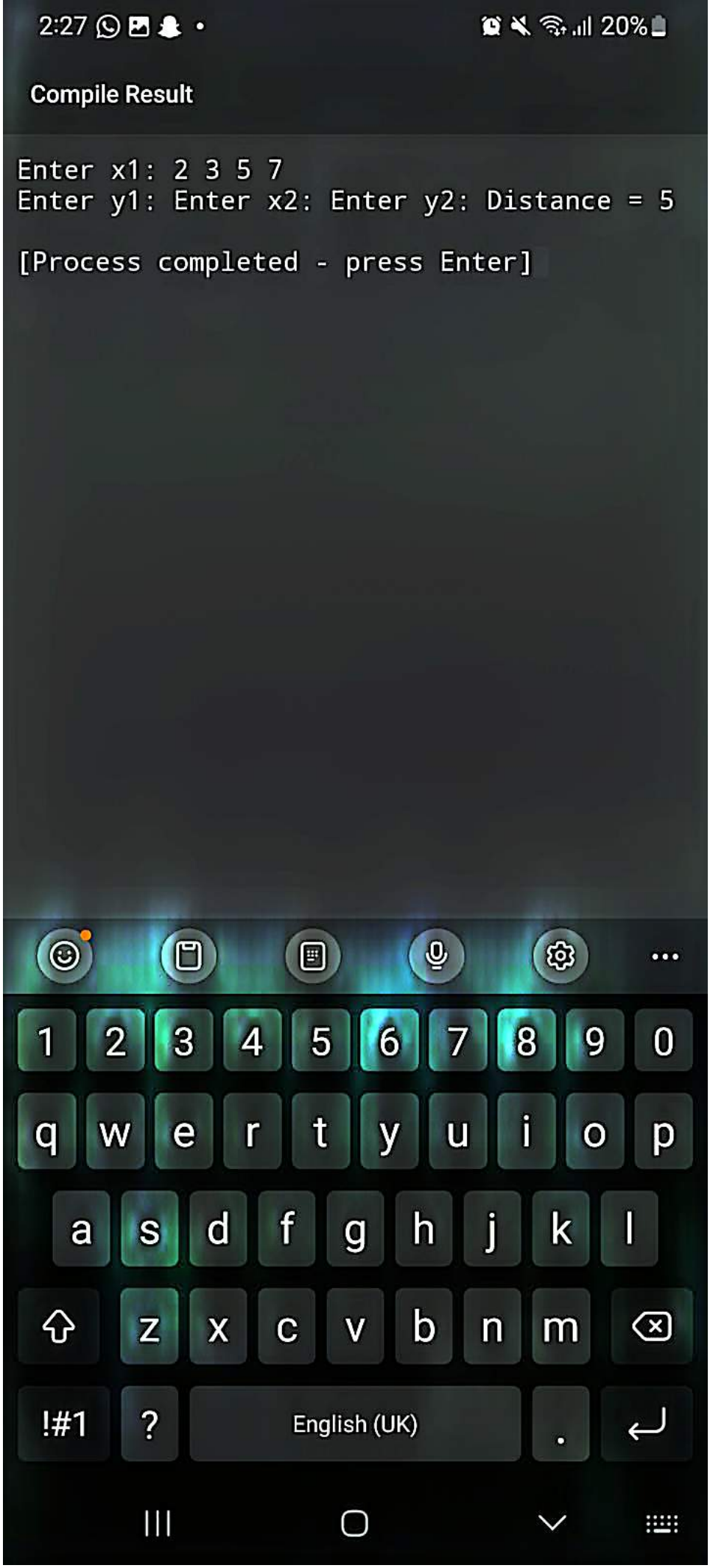
```
# include <iostream>
# include <cmath>
Using namespace std;
   int main ( )
   {
double x1, y1, x2, y2, distance;
cout << " Enter x1 and y1 : ";
 cin >> x1 >> y1 ;
cout << " Enter x2 and y2 :";
 cin >> x2 >> y2;
distance = sqrt (Pow (x2 - x1, 2) + Pow (y2 - y1, 2));
 cout << " Distance = " << distance << end;
     return 0;
   }
```

OUTPUT:

```
Enter x1 and y1 : 2  3
Enter x2 and y2 : 5  7
  Distance = 5
```


2:28
20%
Coding C++
Auto saved at 02:27:39
RUN
MENU
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
double x1, y1, x2, y2;
double distance;
cout << "Enter x1: ";
cin >> x1;
cout << "Enter y1: ";
cin >> y1;
cout << "Enter x2: ";
cin >> x2;
cout << "Enter y2: ";
cin >> y2;
distance = sqrt((x2 - x1) * (x2 - x1) +
(y2 - y1) * (y2 - y1));
cout << "Distance = " << distance << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner


2:27
20%
Compile Result
Enter x1: 2 3 5 7
Enter y1: Enter x2: Enter y2: Distance = 5
[Process completed - press Enter]
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
!#1 ?
English (UK)
.

CamScanner

TASK 05:

Formula : Area of Rectangle

Area = length x Width

```
# include <iostream>
using namespace std;
int main ( )
{
   double length, width, area a;
   cout << Enter length : ";
   cin >> length;
   cout << Enter width: ";
   cin >>width ;
   area = length * width;
   cout << Area of Rectangle
   = "
   <<area << endl;
   return 0;
}
```

AREA OF CIRCLE:

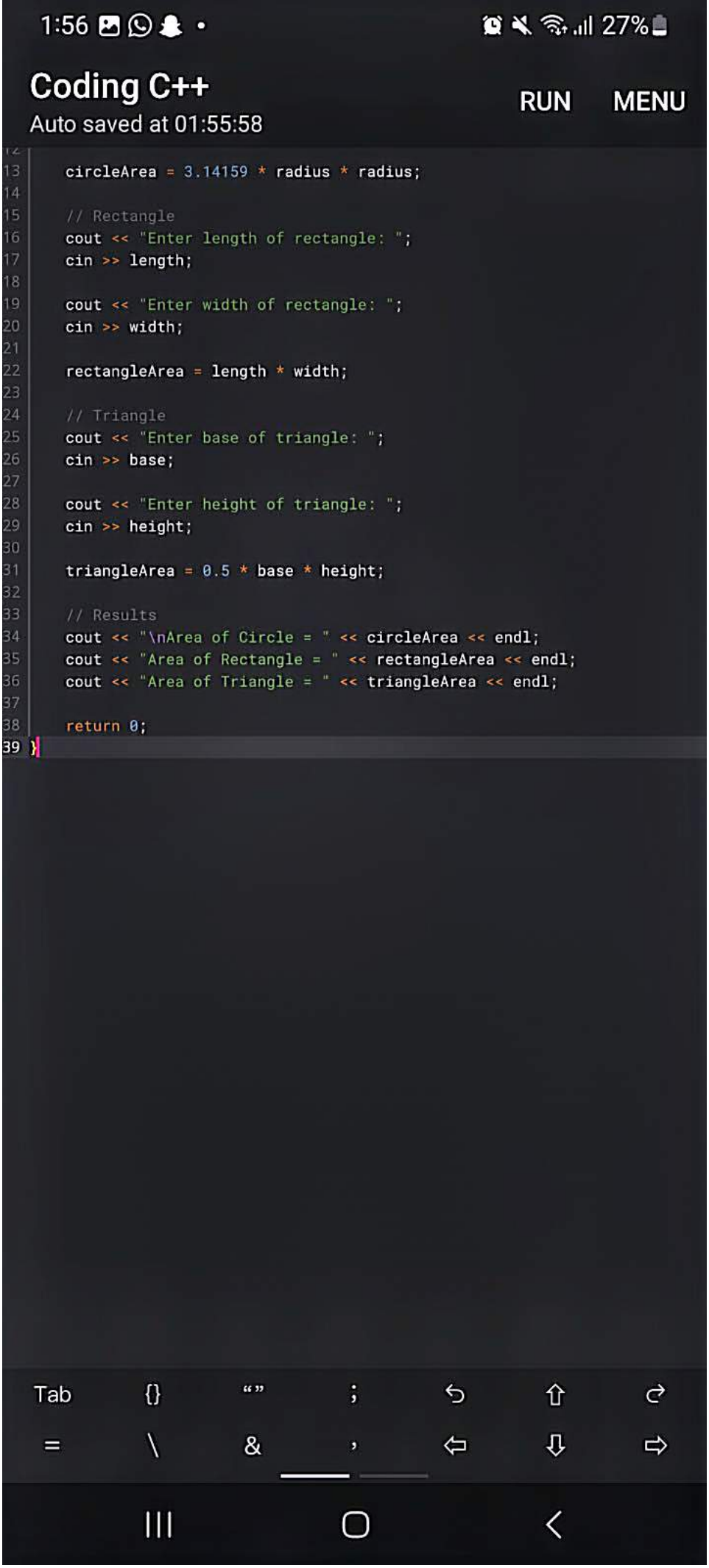
Formula $A = PI \times r^2$

```
#include <iostream>
using namespace std;
    int main ( )
{
   double r, area;
cout double PI = 3.14159;
cout << " Enter radius : ";
 cin >> r;
area = PI * r * r;
cout << Area of circle = " << area << endl;
        return 0;
     }
```

AREA OF TRIANGLE:

Area = Base x height x 0.5

```
# include <iostream>
 using namespace std;
      int main ( )
{
   double height , base, area;
  cout << Enter Base : ";
  cin  >> base ;
 cout << Enter height: ";
  cin >> height ;
 area  =  base * height * 0.5 ;
 cout << Area of triangle = "
          return 0;
       }
```


1:56
27%
Coding C++
Auto saved at 01:55:58
RUN
MENU
circleArea = 3.14159 * radius * radius;
// Rectangle
cout << "Enter length of rectangle: ";
cin >> length;
cout << "Enter width of rectangle: ";
cin >> width;
rectangleArea = length * width;
// Triangle
cout << "Enter base of triangle: ";
cin >> base;
cout << "Enter height of triangle: ";
cin >> height;
triangleArea = 0.5 * base * height;
// Results
cout << "\nArea of Circle = " << circleArea << endl;
cout << "Area of Rectangle = " << rectangleArea << endl;
cout << "Area of Triangle = " << triangleArea << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

```
Enter radius of circle: 5 10 4 8 6
Enter length of rectangle: Enter width of r
ectangle: Enter base of triangle: Enter hei
ght of triangle:
Area of Circle = 78.5397
Area of Rectangle = 40
Area of Triangle = 24

[Process completed - press Enter]
```

CamScanner

2:35 18%

Coding C++

Auto saved at 02:35:54

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    double radius, length, width, base, height;
    double circleArea, rectangleArea, triangleArea;

    // Circle
    cout << "Enter radius of circle: ";
    cin >> radius;

    circleArea = 3.14159 * radius * radius;

    // Rectangle
    cout << "Enter length of rectangle: ";
    cin >> length;

    cout << "Enter width of rectangle: ";
    cin >> width;

    rectangleArea = length * width;

    // Triangle
    cout << "Enter base of triangle: ";
    cin >> base;

    cout << "Enter height of triangle: ";
    cin >> height;

    triangleArea = 0.5 * base * height;

    // Results
    cout << "\nArea of Circle = " << circleArea << endl;
    cout << "Area of Rectangle = " << rectangleArea << endl;
    cout << "Area of Triangle = " << triangleArea << endl;

    return 0;
}
```

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TASK 06:

Total Purchase

Prices:

Item 1 = $12.95
Item 2 = $24.95
Item 3 = $6.95
Item 4 = $14.95
Item 5 = $3.95

Sales tax = 6%

```
# include <iostream>
# include <iomanip>
Using namespace std;
  int main ( )
{
  double item 1 = 12.95;
  double item 2 = 24.95;
  double item 3 = 6.95;
  double item 4 = 14.95;
  double item 5 = 3.95;
    double subtotal;
    double salestax;
    double total;
  subtotal = item 1 + item 2 + item 3 + item 4 + item 5;
    salestax = subtotal + salestax;
    cout << fixed << setPrecision (2);
  cout << " Item 1 ; $ " << item 1 << endl;
  cout << Item 2 ; $ " << item 2 << endl;
  cout << Item 3; $ " << item 3 << endl;
  cout << Item 4, $ " << item 4 << endl;
  cout << Item 5, $ " << item 5 << endl;
  cout << "subtotal : $ " <<subtotal << endl;
  cout << salestax : $ " << salestax << endl;
  cout << "Total: $ " << Total << endl;
    return 0;
}
```

OUTPUT:

```
Item 1 : $ 12.95
Item 2 : $ 24.95
Item 3 : $ 6.95
Item 4 : $ 14.95
Item 5 : $ 3.95
Subtotal : $ 63.75
Salestax : $ 3.83
Total : $ 67.58
```

1:57 26%

Coding C++

Auto saved at 01:57:35

RUN MENU

```
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal;
    double salesTax;
    double total;

    subtotal = item1 + item2 + item3 + item4 + item5;
    salesTax = subtotal * 0.06;
    total = subtotal + salesTax;

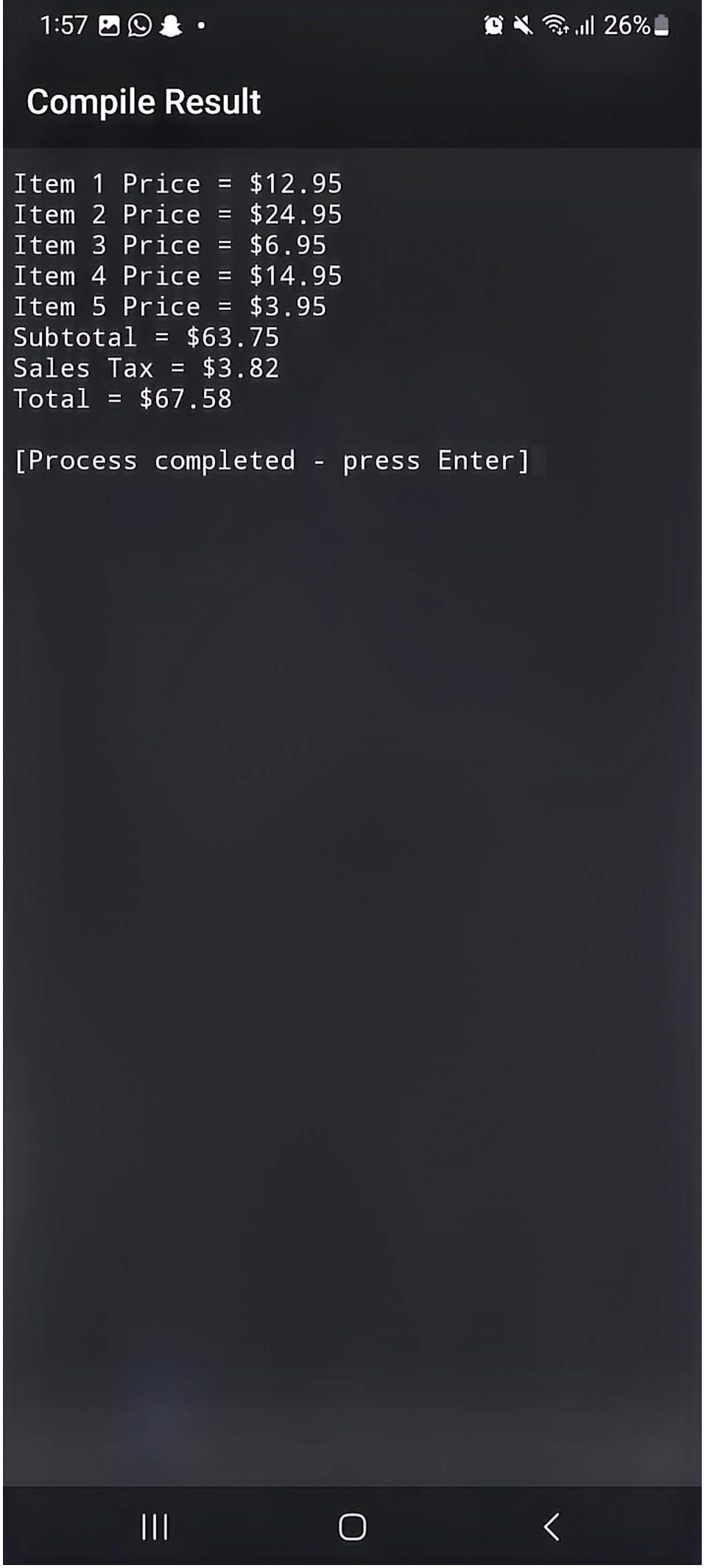
    cout << fixed << setprecision(2);

    cout << "Item 1 Price = $" << item1 << endl;
    cout << "Item 2 Price = $" << item2 << endl;
    cout << "Item 3 Price = $" << item3 << endl;
    cout << "Item 4 Price = $" << item4 << endl;
    cout << "Item 5 Price = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << salesTax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

CamScanner


1:57
26%
Compile Result
Item 1 Price = $12.95
Item 2 Price = $24.95
Item 3 Price = $6.95
Item 4 Price = $14.95
Item 5 Price = $3.95
Subtotal = $63.75
Sales Tax = $3.82
Total = $67.58
[Process completed - press Enter]

CamScanner

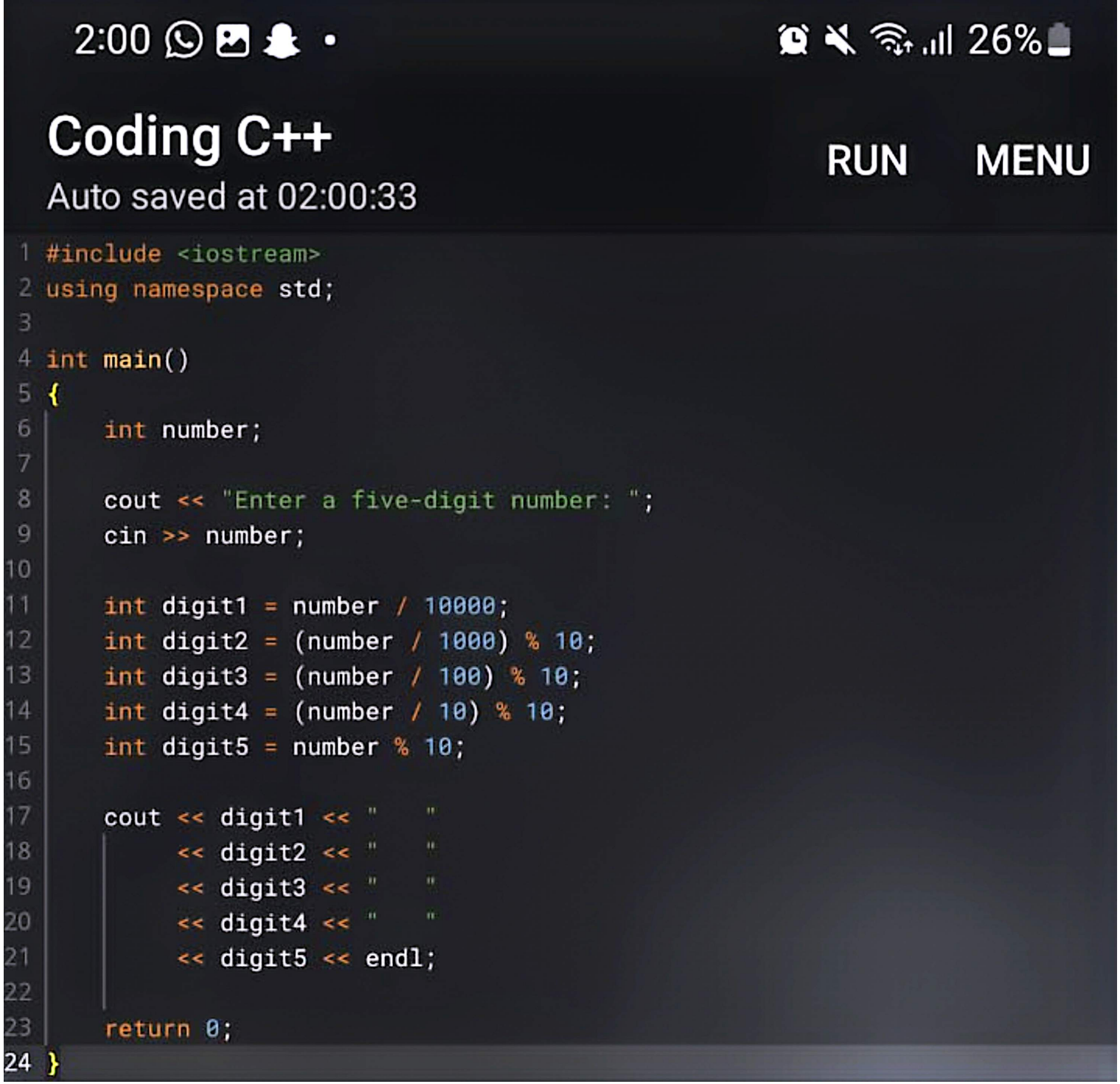
BONUS TASK:

PROGRAM

```
#include <iostream>
using namespace std;
int main ( ) {
        int number;
    cout << "Enter a five-digit integer: ";
        cin >> number;
    cout << number / 10000 << "    "
    "    "   << (number / 1000) % 10 <<
        << (number /100) % 10 <<
    "    "
        << (number/10) % 10 << "
"
        <<number % 10 << endl;
    return 0;
}
```

OUTPUT:

```
Enter a five-digit number : 12345
- 1 2 3 4 - 5
```


2:00
26%
Coding C++
Auto saved at 02:00:33
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int number;
cout << "Enter a five-digit number: ";
cin >> number;
int digit1 = number / 10000;
int digit2 = (number / 1000) % 10;
int digit3 = (number / 100) % 10;
int digit4 = (number / 10) % 10;
int digit5 = number % 10;
cout << digit1 << " "
<< digit2 << " "
<< digit3 << " "
<< digit4 << " "
<< digit5 << endl;
return 0;
}

CamScanner

2:00 26%

Compile Result

```
Enter a five-digit number: 12345
1   2   3   4   5

[Process completed - press Enter]
```

CamScanner

HOME TASKS:

01:

PROGRAM

```
# include <iostream>
using namespace std;
int main ( ) {
    int var1, var2, temp;
  cout << Enter value of var1:";
   cin >> var1;
  cout << Enter value of var2:";
   cin >> var2;
  cout << " Before swapping:" << end1;
```

```
cout << " var1 = " << var1 << endl;
cout << "var 2 = " << var2 << endl;

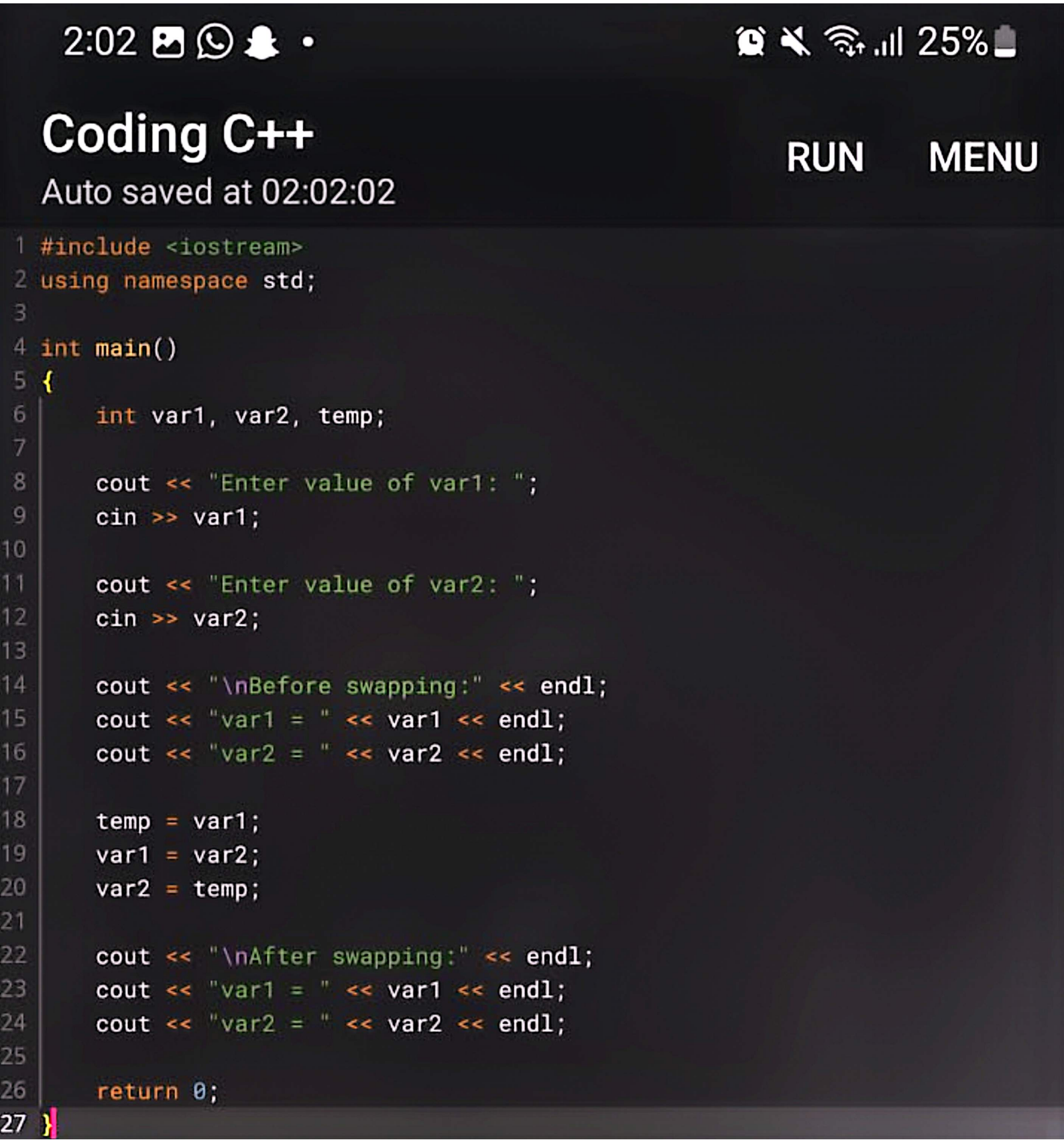
    // Swapping values
    temp = var1;
    var1 = var2;
    var2 = temp;
cout << "After swapping:" << endl;
cout << " var 1 = " << var1 << endl;
cout << " var2 = " << var2 << endl;

    return 0;
}
```

OUTPUT:

```
Enter value of var1: 10
Enter value of var2: 20
    After swapping
var 1 = 20
var 2 = 10
```

CamScanner


2:02
25%
Coding C++
Auto saved at 02:02:02
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int var1, var2, temp;
cout << "Enter value of var1: ";
cin >> var1;
cout << "Enter value of var2: ";
cin >> var2;
cout << "\nBefore swapping:" << endl;
cout << "var1 = " << var1 << endl;
cout << "var2 = " << var2 << endl;
temp = var1;
var1 = var2;
var2 = temp;
cout << "\nAfter swapping:" << endl;
cout << "var1 = " << var1 << endl;
cout << "var2 = " << var2 << endl;
return 0;
}

CamScanner

```
Enter value of var1: 10 20
Enter value of var2:
Before swapping:
var1 = 10
var2 = 20

After swapping:
var1 = 20
var2 = 10

[Process completed - press Enter]
```

CamScanner

TASK 2:

Program

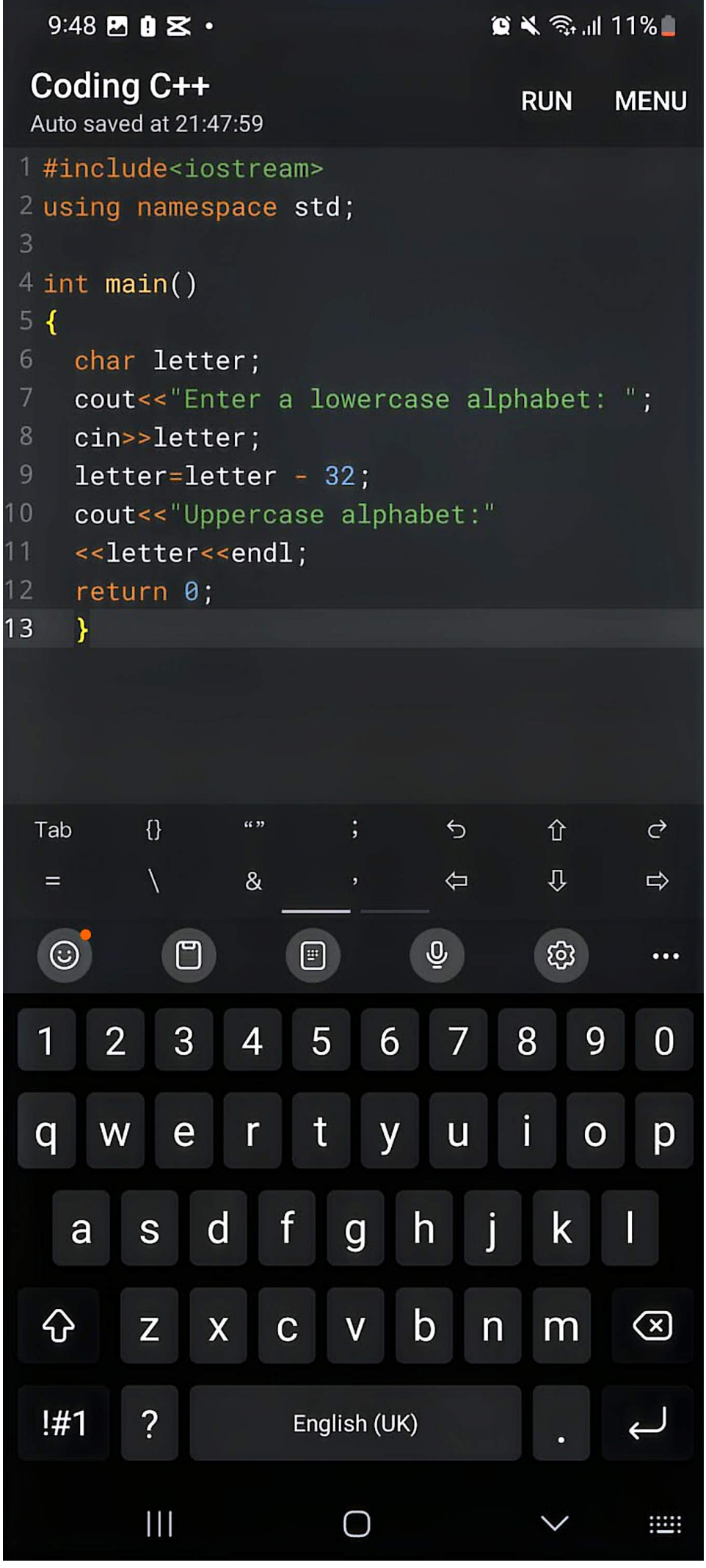
```
#include <iostream>
using namespace std;
int main ( ) {
    char letter;
cout << "Enter a lowercase alphabet: ";
cin >> letter;

    letter = letter - 32;
cout << "Uppercase alphabet: " << letter << endl;

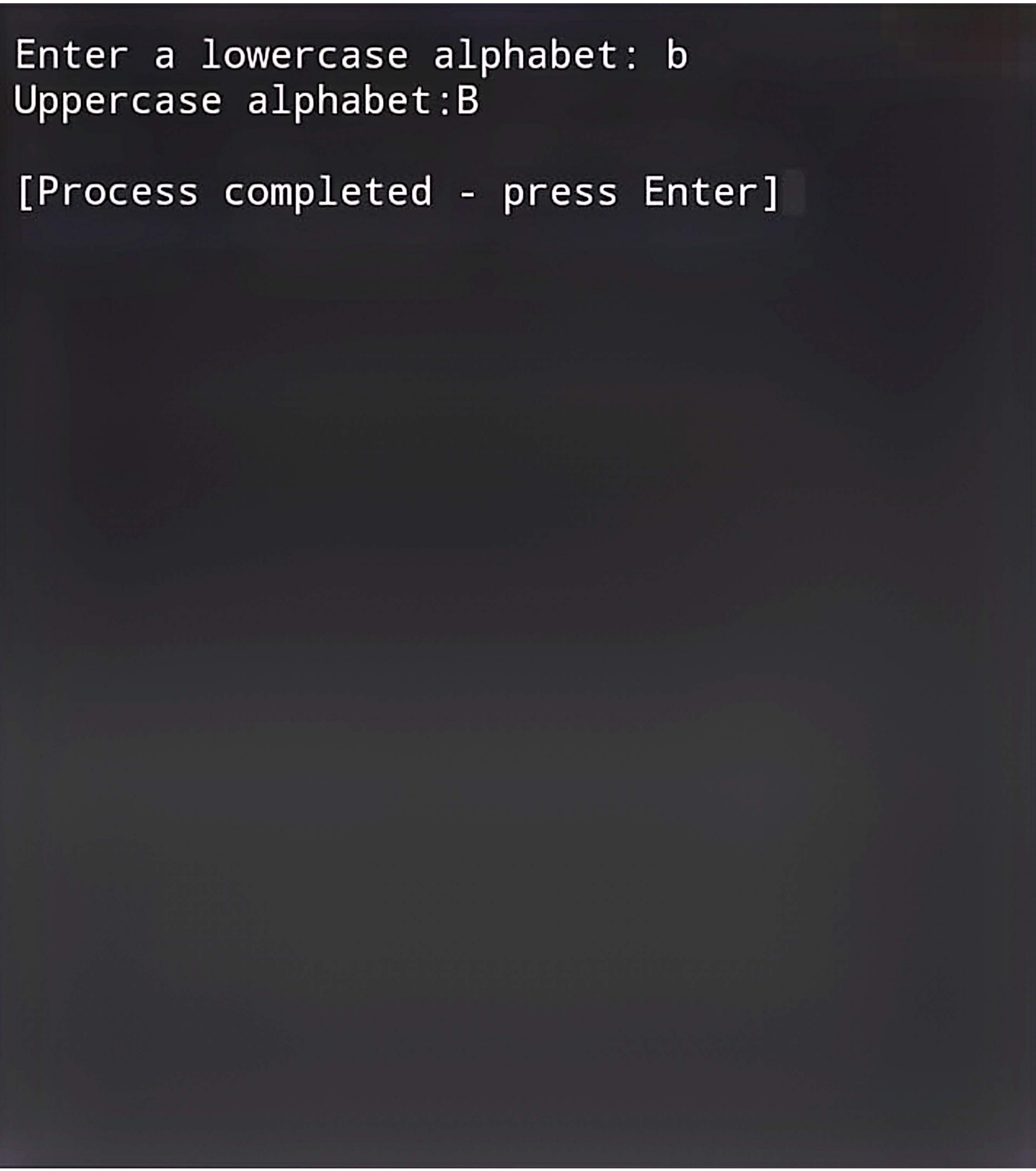
    return 0;
}
```

OUTPUT:

```
Enter a lowecase alphabet = b
Uppercase alphabet = B
```


9:48
11%
Coding C++
Auto saved at 21:47:59
RUN
MENU
#include<iostream>
using namespace std;
int main()
{
char letter;
cout<<"Enter a lowercase alphabet: ";
cin>>letter;
letter=letter - 32;
cout<<"Uppercase alphabet:"
<<letter<<endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,
English (UK)
!#1
?
.

CamScanner


Enter a lowercase alphabet: b
Uppercase alphabet:B
[Process completed - press Enter]

CamScanner